ÖZYEĞİN UNIVERSITY
Math 104 – Fall 2017

Midterm 1

Seat Number: .....

Name: .....

Number: .....

- Please read all directions and sign the honor code below.
- The exam will be 110 minutes long and consists of 6 questions in 7 pages. Make sure you have all the pages.
- Be sure to write your exam seat number, full name, and student number above.
- Calculators and other electronic devices such as smart phones are not allowed.
- Leaving the exam room temporarily during the exam is not allowed.
- You cannot leave the exam during the first 35 minutes.
- Please raise your hand quietly if you have any questions for the instructors or for the proctors.
- Answers given without supporting work will receive no credit.

Honor Code: I pledge under my honor that I **did not give or receive any unauthorized assistance** in this work.

Signed:.....

Question	Max. Points	Score
1	16	
2	18	
3	16	
4	16	
5	18	
6	16	
Total	100	

Q1) [$8 + 8 = 16$ pts] **(Calculus with Parametric Curves)** Given the parametric curve

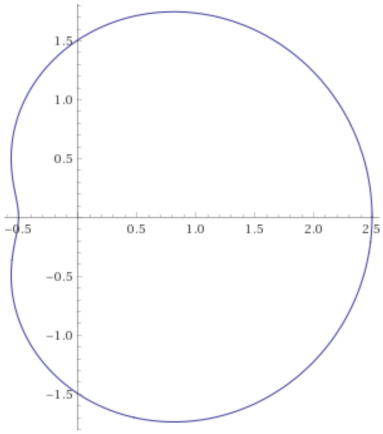
$$x = \sec t, \quad y = \tan t,$$

(a) Find an equation for the tangent line to the curve at $t = \dfrac{\pi}{6}$.

(b) Find the value of $\dfrac{d^2y}{dx^2}$ at $t = \dfrac{\pi}{6}$.

Q2) [$10 + 8 = 18$ pts] **(Areas and Lengths in Polar Coordinates)**

(a) Find the area of the region inside the *limaçon* $r = \dfrac{3}{2} + \cos\theta$. $\left(\text{Hint: } \cos^2\theta = \dfrac{1+\cos 2\theta}{2}\right)$



Graph of $r = \dfrac{3}{2} + \cos\theta$.

(b) Find the length of the polar curve $r = e^{\theta}$ from $\theta = 0$ to $\theta = \pi$.

Q3) [$4 \times 4 = 16$ pts] **(Vectors and Geometry of Space)** Given the vectors $\mathbf{u} = \mathbf{i} - \mathbf{j}$, and $\mathbf{v} = 3\mathbf{i} - \mathbf{j} + 5\mathbf{k}$,

(a) Find $|\mathbf{u}|$, $|\mathbf{v}|$, and $\mathbf{u} \cdot \mathbf{v}$.

(b) Find the cosine of the angle between $\mathbf{u}$ and $\mathbf{v}$.

(c) Find the vector projection of $\mathbf{u}$ onto $\mathbf{v}$.

(d) Find the cross product $\mathbf{v} \times \mathbf{u}$.

Q4) [$8 + 8 = 16$ pts] **(Lines and Planes in Space)**

(a) Find the plane containing the two intersecting lines parametrized as below.

$$x = -1 + t, \qquad y = 2 + t, \qquad z = 1 - t, \qquad -\infty < t < \infty.$$
$$x = 1 - 4s, \qquad y = 1 + 2s, \qquad z = 2 - 2s, \qquad -\infty < s < \infty.$$

(b) Find the distance from the point $S(1, 4, 0)$ to the plane through the points $P(0, 0, 0)$, $Q(2, 0, -1)$, and $R(2, -1, 0)$.

Q5) [$3 \times 6 = 18$ pts] **(Vector-Valued Functions and Motion in Space)** Consider the motion of a particle in space in which the velocity of the particle at any time t is given by the vector function

$$\mathbf{v}(t) = (t^3 + t)\,\mathbf{i} + 2\sqrt{2}t\,\mathbf{j} + 2t^2\,\mathbf{k}.$$

(a) Find the particle's speed, and the particle's acceleration vector at time $t = 2$.

(b) Find $\mathbf{r}(t)$, the position of the particle as a function of time, if the initial position the particle is $\mathbf{r}(0) = \mathbf{i} + \mathbf{j}$.

(c) Find the arc length parameter $s(t)$ along the curve $\mathbf{r}(t)$ with the base point at $t = 0$, for nonnegative values of t.

Q6) [$4 \times 4 = 16$ pts] **(Vector-Valued Functions and Motion in Space)** For the curve given by the vector function $\mathbf{r}(t) = (\sin t)\mathbf{i} + (\sqrt{2}\cos t)\mathbf{j} + (\sin t)\mathbf{k}$,

(a) Find its unit tangent vector $\mathbf{T}$ at $t = \dfrac{\pi}{4}$.

(b) Find its curvature κ.

(c) Find its principal normal vector $\mathbf{N}$ at $t = \dfrac{\pi}{4}$.

(d) Find the length of the portion of the curve when $0 \leq t \leq 2\pi$.